Signed baby-block evaluator for the calibrated first response

Public G,Q -> signed baby / giant tables

b+n addition calls and b+1+n point slots per generator; exact final-block clipping

Plus/minus factors -> first spectral jet

(L-J+s)(L+J+s) = (L^2-J^2) + s*(2L); preserve relative y signs

Baby coordinates -> norm algebra

H(Z), beta=x(jQ), gamma=y(jG)y(jQ); norm(E0+s E1)/norm(D^2)

Singular determinant -> calibrated point action

Unit-only pivots retain B at zero constant; one calibration and seven shifts decode

Construction qualified 2026-11-08; reconciled and packaged 2026-11-09, T-1945. The B-only spectral evaluator now builds signed baby/giant point tables of square-root size and contracts paired factors without constructing the complete r-point target table. It also has an explicit signed baby-coordinate algebra, with a unit-pivot determinant reference that preserves the first coefficient at a marked zero. The mathematical proof and native checks remain unverified producer evidence. [pcc:unverified]

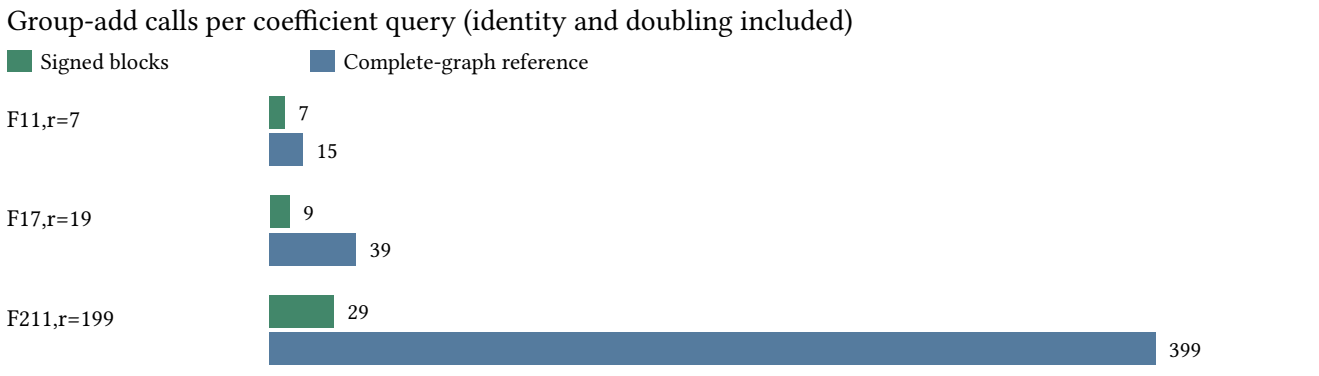
Paired first response. Write x(A+T)=H_A-K_A and x(A-T)=H_A+K_A, with the standard signed addition expressions. For matched source/target babies put L=H_A-H_B and J=K_A-K_B. The two factors contract to (L^2-J^2)+s*(2L), modulo s^2. A streaming (A,B) multiplication needs no C and never divides by a zero norm factor. A wrong relative sign changes the constant factor by 4*K_A*K_B; the native checker rejects that omission in every executed pair. These are factor controls, not an assertion that complete zero-order norms distinguish Q and -Q. [pcc:unverified]

Public constructor. Use b=max(1,floor(sqrt(r))/2), m=2b+1, n=ceil(r/m), with giant centers b+i*m and babies -b through b. Exact clipping retains each index0..r-1 once, including a final block whose center may lie beyond r-1. For each supplied generator, constructing babies and giants takes b+n addition calls and b+1+n stored point slots. The G table is reusable; Q tables change with the target. The qualification harness also constructs full graphs for its references, so the component storage figure is not total process memory. [pcc:unverified]

Signed norm algebra. The source baby polynomial H has roots x(jG), j=1..b. Two interpolants supply beta(x(jG))=x(jQ) and gamma(x(jG))=y(jG)*y(jQ). Gamma retains the relative sign. The proof gives a paired numerator E0+s*E1 over K[Z]/H and a nonzero denominator norm. Native multiplication matrices and full row/column pivoting compute this norm using only pivots with nonzero constant terms; the last zero constant pivot is multiplied without inversion. The dense reference is fully charged rather than assigned a fast resultant cost. [pcc:unverified]

Complete decoder panel. All381 declared targets/inputs recover using the calibrated seven-shift decoder:28 on F11/r7,304 on F17/r19 and49 on F211/r199. The2,694 coefficient queries comprise27 reusable calibrations and2,667 shifted queries. They cover111,786 underlying factors as47,258 pairs and17,270 singletons. All47,258 wrong-sign pair controls reject. The signed algebra agrees on11,382 complete paired blocks, including2,190 with a zero constant product. The candidate reads public points and produces the response before the known-label scalar check. [pcc:unverified]

Exact point-call costs. For r7,r19,r199, candidate constructor calls are4,6,21 versus7,19,199 for the complete graph reference; candidate point slots are5,7,22. A query uses7,9,29 group-add calls respectively, versus15,39,399 for the complete-graph reference. These counters include identity and doubling calls. They describe the named implementations and are not timings or an end-to-end speedup. [pcc:unverified]



Source: response.log. Per query; setup, field operations, decoding and references separately retained.

Field costs retained. On the frozen first shifted query (Q=2G,U=O,i=0), candidate quadratic-field multiplication/inversion counts are242/10,636/24,8730/218; complete-reference counts are213/12,705/36,12477/396. Thus the smallest panel uses more candidate multiplications even while its point-call count falls. The dense algebra reference separately costs521/23,1930/68,91595/1663. Candidate counters retain sign guards and block recombination checks; they exclude the separately counted interior determinant work. Inversions in the small-field backend include their exponentiation multiplications. Interpolation setup, cubic decoding, fixture creation, reference graphs and fresh replays remain additional costs. Raw per-query receipts and whole-qualifier totals are preserved. [pcc:unverified]

P-192 component. Native full-width quadratic arithmetic checks24 prescribed27-factor blocks on the existing public Q, mu_G and relation_lambda_G. The layout is b4,m9,n3; source and target builders each use7 addition calls and8 point slots. Every paired first coefficient agrees with direct27-factor prefix/suffix contraction, and288 wrong-sign pair controls reject. The supplied U=[j](G-Q) deliberately puts the unique zero in the selected block. The complete P192 layout is computed as integers without allocation; no full P192 B response or arbitrary-input decoder is claimed by these subset checks. Its counters count explicit K multiplication/inverse calls, excluding inverse internals, and are not mixed with the small-field counter model. [pcc:unverified]

Proof, replay and actual failures. Source and scope were published before affected data. The initial build failed because two unused generic polynomial locals lacked type inference; their removal was published before qualification, and both logs remain. Completed fresh-copy science outputs and both operation receipts reproduce byte-for-byte. Changed small and P192 coefficient records reject with101. The required legacy library check timed out compiling isogeny_algos after120 seconds, exit124; it is not a publication-head pass. The interrupted process is terminal and was not restarted after reconciliation. No cloud experiment, managed generator or old freeze lane was started or modified. [pcc:unverified]

Next field-contraction construction. The current stream still contracts a linear number of field factors; its Lagrange baby interpolation is quadratic in b and the dense algebra norm is cubic per block. The signed degree-b presentation now makes the next backend precise: compute the first coefficient using field subresultants with a separate simple-common-root branch, then investigate sharing across giant blocks. The acceptance obligation is equality at both unit and singular blocks with explicit setup and field costs. Reducing point generation does not supply that contraction step. [pcc:unverified]

Sources checked. Live Round60 signed-pair and Round62 calibrated-decoder proofs are inherited, with hashes. Bernstein-De Feo-Leroux-Smith, https://arxiv.org/pdf/2003.10118, Lemma4.3 and4.9 provide one-curve pairing/resultant background, Example4.13 gives jet evaluation, and section4.16 identifies extra-coordinate issues for products of elliptic curves. Our retained beta,gamma algebra addresses the representation contract without importing the paper's one-curve runtime. KB tools and the local wrapper are unavailable; no novelty or prior-art absence claim follows. Current AGENTS additions on persistent research and counter pricing were read; this study emits exact primitive counts and does not use ecbench table-oracle price bands. [pcc:unverified]